

Recurrence, retirement, and a comeback: 1997-2026

LSTM
(Hochreiter & Schmidhuber)

1997

"Attention Is All
You Need" (Vaswani et al.)

2017

xLSTM (Beck et al.)
Hochreiter, senior author

2024

*2026: RNNs still earn their keep in streaming/edge
and small-data settings*

seq2seq (Sutskever et al.)
GRU (Cho et al.)

2014

Mamba (Gu & Dao)
state-space models return

2023

